Embedded Peer Financing in Tokenized Marketplaces

Many digital marketplaces are expanding beyond matching buyers and sellers by embedding financing directly into transactions. Can financing provided by marketplace participants themselves increase trade? While financing may stimulate demand by relaxing liquidity constraints, it may also divert capital toward lending and alter supply decisions. We study this question in a tokenized marketplace where users can simultaneously act as buyers, sellers, and lenders. We examine the staggered introduction of peer-to-peer financing on Blur, a leading marketplace for non-fungible tokens (NFTs). Using transaction-level data from Blur and its largest competitor, OpenSea, we estimate the impact of financing using Synthetic Difference-in-Differences and triple-difference designs. We find that financing increases quantities sold by 49% without significantly affecting average prices. Financing has little effect on activity on the rival platform, suggesting limited cross-platform spillovers. Financing primarily increases activity among existing users rather than attracting new buyers or sellers, although it attracts new participants on the lending side. In addition, financing shifts demand toward higher-value assets. Our findings show that participant-provided financing can substantially increase marketplace activity despite competing incentives to allocate capital between lending and trading. More broadly, the results suggest that tokenization enables platforms to integrate exchange and capital provision within the same marketplace, allowing participants to simultaneously act as buyers, sellers, and lenders. In doing so, embedded peer financing can deepen marketplace engagement and reshape the allocation of capital and trade within digital ecosystems.

Key words: Digital Platforms, Peer-to-Peer Lending, Tokenization

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1. Introduction

Digital assets based on the Web3 ecosystem are expected to reach a market capitalization exceeding USD 2 trillion by 2030 (McKinsey & Company 2024).[1] Platforms in this ecosystem operate across diverse markets from digital collectibles (Blur, OpenSea) to tokenized real-world assets such as real estate (RealT) and securities (Securitize). As these platforms evolve, they increasingly perform functions beyond matching buyers and sellers. In particular, tokenization allows ownership claims to be represented as blockchain-based tokens that can be transferred, traded, and pledged as collateral through smart contracts. Tokenization thus allows financing to be embedded directly into marketplace activity, enabling platforms to facilitate not only trade but also capital formation among participants. This raises an important question: how does enabling participants to simultaneously trade and provide capital affect user behavior and marketplace activity?

Tokenization has significant potential to transform financial intermediation by relaxing liquidity constraints and enabling new forms of value creation (World Economic Forum 2025, Fink and Goldstein 2025). This potential stems from two key properties: fractional ownership, which broadens participation in asset markets, and programmable ownership, which enables smart contracts to automatically transfer ownership when pre-specified conditions are met.[2] The latter makes collateralized lending possible with minimal reliance on traditional intermediaries or legal enforcement, reducing the costs of asset-backed lending. As a result, platforms such as RealT and Centrifuge (real estate), Ondo Finance and Securitize (securities), and Paxos (commodities) already allow tokenized assets to serve as collateral for smart-contract-enabled loans.

The introduction of financing into tokenized marketplaces may give rise to competing economic effects. On the one hand, financing can relax liquidity constraints and enable users to acquire assets that would otherwise remain beyond their immediate purchasing capacity. Prior research on consumer credit and buy-now-pay-later programs (e.g., Akana and Doubinko 2023) suggests that access to financing can stimulate demand and increase transaction activity. On the other hand, when financing is supplied by marketplace participants themselves, capital that might otherwise be used for purchases may instead be allocated toward lending. Financing may also induce supply-side responses by encouraging sellers to list more assets for sale. Consequently, the overall impact of participant-provided financing on marketplace activity, platform growth, and user participation remains unclear.

[1] Web3 refers to a decentralized internet infrastructure built on blockchain technology, where users transact and interact through distributed protocols rather than centralized intermediaries with programmable incentive structures based on smart contracts.

[2] All parties agree on the rules in advance, which are encoded as smart contracts on an immutable blockchain. The token's smart-contract metadata records ownership, and transfers ownership automatically upon sale or loan default according to the contract's terms.

To study these questions, we examine the introduction of peer-to-peer financing in an NFT marketplace, a setting that provides a unique opportunity to observe embedded finance in a tokenized market. NFTs represent ownership of digital assets such as artwork, music, and virtual collectibles. These assets are typically organized into collections that share a common creator, brand, or theme. For example, the Bored Ape Yacht Club (BAYC), one of the most prominent NFT collections, contains more than 10,000 unique digital collectibles that trade across multiple marketplaces. During our study period, the two largest NFT marketplaces, Blur and OpenSea, accounted for approximately 90% of market activity.

Our focal platform, Blur, introduced peer-to-peer asset-backed financing in a staggered manner across leading NFT collections in 2023. Through this feature, marketplace participants could provide loans to one another using NFTs as collateral. Each loan was tied to a specific NFT collection selected by the lender, and borrowers could use the funds only to purchase NFTs from that collection.[3] This financing mechanism differs from traditional buy-now-pay-later programs in two important respects. First, the loans are collateralized by the tokenized asset itself rather than being unsecured. Second, the platform bears no credit or counterparty risk; financing is provided peer-to-peer by users acting as lenders. The platform's role is to enable financing and implement the technological infrastructure.

We investigate how embedded peer financing affects marketplace outcomes. In particular, we examine its impact on transaction volume, prices, and user participation, including the entry of new users and changes in activity among existing buyers and sellers. The institutional features of NFT marketplaces allow us to study several pathways through which financing may affect marketplace outcomes, including cross-platform spillovers, heterogeneous responses across users and assets, and the relative contributions of demand- and supply-side responses.[4] Understanding these mechanisms is important because financing is becoming an increasingly common feature of tokenized marketplaces, yet little is known about how it shapes market activity and participant behavior.

We obtain transaction-level data from January to December 2023 from both Blur and OpenSea.[5] The transaction records include NFT characteristics, buyer and seller identifiers, transaction prices, and indicators of whether a purchase was financed. Our data includes 49 NFT collections, accounting for over 66% of sales on Blur.[6] To evaluate the impact of financing, we compare collections on

[3] Figure 6, in the Appendix provides an example NFT from the BAYC collection together with the financing contracts used to fund purchases within the collection.

[4] Throughout the paper, "supply" refers to the NFTs listed for sale in a given period, not the number of NFTs in a collection. While the total number of NFTs in a collection (e.g., 10,000 for BAYC) is determined at creation and cannot change, the share of those NFTs that owners choose to list for sale varies over time.

[5] NFTs are interoperable across platforms, i.e., the same asset can be listed and traded on any marketplace.

[6] Note that we collected data for about 200 top collections, but restricted our sample to the 49 collections to obtain a balanced sample; some of the other collections are used to create an alternate control group.

the focal platform where financing was enabled (treated collections) with collections on the same platform where financing was not available (potential control collections). We exclude wash trades from our sample by following the wash-trade detection methodology developed by Dune Analytics, which is widely used by digital asset platforms and in empirical work on NFT markets.[7]

Given the observational nature of our study, identification requires addressing several considerations. First, the focal platform could have chosen the treated collections based on popularity or similar factors. Second, because financing was rolled out across collections at different times, we need to appropriately accommodate this staggered timing. Third, users could potentially substitute assets between treatment and control collections in their buying (or selling) decisions. A user might purchase an asset from a treated collection, and due to a budget constraint, forego purchasing an asset from a control collection. For valid inference, we typically need the Stable Unit Treatment Value Assumption (SUTVA), which holds when only the treated collection is impacted by financing, but not the control collection. Therefore, such a substitution would be a threat to causal inference.

We address these considerations as follows. First, we use the Synthetic Difference-in-Differences (Synthetic DiD) estimator (Arkhangelsky et al. 2021), which is widely applied in recent empirical research, including studies of online creators (Qian and Xie 2026), music playlists (Pachali and Datta 2025), and advertising (Lambrecht et al. 2024). It constructs a weighted combination of control collections that matches the pre-treatment trajectory of each treated collection, accommodating the possibility that no single control collection serves as a perfect counterpart, and we apply the weighting and differencing cohort-by-cohort to accommodate staggered timing.

Second, and central to our identification, we exploit the interoperability of NFTs to estimate a triple-difference (DDD) specification that compares treated and control collections, before and after financing, and across Blur and OpenSea. Because the same collections are traded on both platforms while financing is available only on Blur, any collection-specific demand shock that might confound the treated-versus-control comparison appears on both platforms and is differenced out, isolating the Blur-specific effect of financing (Olden and Møen 2022). This design addresses the concern that financing was enabled for collections experiencing unobserved changes in demand.

Third, because we can identify which transactions used Blur's lending mechanism, we can attribute any effect of financing to its actual use rather than to its mere availability. Fourth, to address SUTVA, we restrict the control group to collections with low user overlap relative to treated collections (similar to Zhang et al. 2026), and we re-estimate our model after eliminating users who overlap between the treatment and control groups. We also report Two-Way Fixed Effects

[7] See https://dune.com/blog/nft-wash-trading-on-ethereum

(TWFE) and Two-Stage DiD (Gardner 2022) estimates, the latter directly accounting for staggered treatment timing.

We find that introducing financing substantially increases marketplace activity on the focal platform. Quantities sold for treated collections on Blur increase by 49%. Despite this large effect on trading activity, we observe a statistically insignificant change in average NFT prices, suggesting that financing primarily affects transaction volume and the composition of assets traded rather than average prices. To better understand the source of these effects, we examine whether the increase in activity is driven by the acquisition of new users or by changes in the behavior of existing participants. We find that the increase in sales is primarily driven by greater trading activity among existing users. Financing does not lead to a significant increase in the number of buyers or sellers on the platform. However, Blur does attract new participants on the lending side, with approximately 30% of lenders being new to the platform. These findings suggest that embedded financing primarily deepens engagement among existing traders while expanding participation through the introduction of a new lender role.

We next examine how financing affects the composition of assets traded. Financing increases the quantities sold of high-end (more expensive or rarer) NFTs while having little effect on low-end NFTs, shifting the mix of sales toward higher-value assets. Finally, we investigate the mechanisms underlying the increase in trading activity by decomposing the effects into demand- and supply-side responses. We characterize the supply-side response as changes in the number of listings and the demand-side response as changes in the probability that a listed NFT is purchased.[8] We find that financing increases both margins. Approximately 39% of the increase in sales can be attributed to a rise in listings, while the remaining 61% is driven by a higher purchase probability for listed NFTs. These findings indicate that financing stimulates activity on both sides of the marketplace, although the majority of the increase in trading volume arises from stronger buyer demand.

Despite these substantial effects on the focal platform, we find little evidence of spillovers to the competing marketplace, i.e., OpenSea. Specifically, financing does not significantly affect quantities or prices on the rival platform OpenSea, even though the underlying assets are portable across marketplaces. The benefits of financing, therefore, appear to remain localized to the platform offering the financing mechanism.

As smart-contract-enabled lending can be implemented across a broad range of tokenized assets, including art, securities, real estate, and commodities, our findings have implications beyond the NFT context. More broadly, they suggest that tokenization enables platforms to integrate financing directly into marketplace activity, blurring the traditional distinction between trading and

[8] Throughout the paper, "supply" refers to the NFTs listed for sale in a given period, not the number of NFTs in a collection.

capital provision by allowing participants to act as buyers, sellers, and lenders within the same ecosystem. We find that these financing mechanisms increase trading activity primarily through greater engagement among existing users and purchases of higher-value assets, rather than through substantial growth in the buyer and seller base.

This paper contributes to the emerging literature on tokenized assets and digital marketplaces (Liang et al. 2026, Zhang et al. 2026). Prior research has emphasized the role of tokenization in expanding participation, overcoming coordination challenges, creating value, and supporting governance in decentralized systems (Cong et al. 2021, Chod et al. 2022, Bakos and Halaburda 2022, Howell et al. 2020, Catalini and Gans 2020, Sockin and Xiong 2023). Much of this work views tokens primarily as incentive, coordination, or governance mechanisms. We complement this perspective by highlighting an additional role of tokenization: enabling capital provision through collateralized peer-to-peer financing. By examining how embedded financing affects marketplace activity, user behavior, and platform outcomes, we show how tokenization can reshape not only how assets are traded, but also how capital is supplied within digital marketplaces.

Our findings have several implications for platform design. As platforms increasingly integrate financial services into transactions, an important question is whether financing functions primarily as a payment convenience, a customer acquisition tool, or a mechanism for increasing engagement within an existing user base. Our findings indicate that participant-provided financing primarily deepens activity among existing users rather than attracting new buyers and sellers, while simultaneously expanding participation through the entry of capital providers. By enabling users to supply as well as consume capital, financing broadens the set of interactions that a platform can facilitate and monetize. More generally, the results suggest that financial infrastructure may serve as a source of platform differentiation even in markets where underlying assets are portable across platforms. By facilitating credit provision among participants rather than intermediating it directly, platforms can expand market functionality while limiting capital requirements and credit-risk exposure.

2. Literature Review

Our paper broadly relates to two streams of literature. First, Web3 platforms and tokenization, including work on digital assets such as NFTs, and second, peer-to-peer lending. Prior work on tokenomics examines how tokens affect participation, value creation and coordination among early adopters (Cong et al. 2021, Chod et al. 2022, Bakos and Halaburda 2022, Howell et al. 2020), and how decentralization and governance mechanisms shape the platform outcomes (Catalini and Gans 2020, Sockin and Xiong 2023). However, this literature largely treats tokens as incentive devices rather than capital provision mechanisms. We build on this work by examining how tokenization enables peer-provided, collateral-backed lending, and how such financing shapes user behavior and platform growth.

Within tokenization, our research also contributes to the nascent yet growing body of work on NFTs. Recent research has explored how features of NFT markets, such as rarity (Yuan et al. 2025), seller experience (Liu et al. 2026), interoperability (Liu and Cui 2025), and the visibility of transactions on the blockchain (Liang et al. 2026), affect trading behavior, sales, and prices. While this research focuses on outcomes within the platform, a separate stream examines spillovers from NFTs to firms, including the effects of NFT introduction and brand extension on offline product sales (Kanellopoulos et al. 2026) and firm performance (Kuzmich et al. 2025, Brandes and Dölp 2025). We build on these streams of research by exploring the impact of financing on platform outcomes and growth. In this sense, our study is related to Zhang et al. (2026), which studies market thickness and platform growth through quality-signaling mechanisms that alter when creators incur the cost of registering assets on the blockchain. In contrast, we examine how participant-provided financing affects trading activity, user engagement, and platform growth.

This paper also contributes to the literature on peer-to-peer lending, which has largely focused on non-tokenized settings. Prior research examines general-purpose credit markets in which lenders and borrowers occupy distinct roles, with platforms mitigating information frictions through screening technologies and soft information (Butler et al. 2017, Balyuk 2023, Vallee and Zeng 2019, Lu et al. 2022). The closest conceptual antecedent is the trade credit literature, which studies financing tied to transactions between trading partners (Petersen and Rajan 1997, Fisman and Love 2004, Cuñat 2007). However, even in these settings, trading and financing functions remain largely separate. In contrast, our setting allows marketplace participants to act simultaneously as buyers, sellers, and lenders, blurring the boundary between trade and capital provision. As a result, participants allocate capital not only through their purchasing decisions but also through their lending decisions, creating competing incentives absent from traditional peer-to-peer lending and trade credit markets. Whether such participant-provided financing expands trade or instead diverts resources away from it remains an open empirical question.

Finally, to the extent that loan offers in our setting are tied to specific collections and cannot be used as credit for purchasing any NFT, financing shares some characteristics with traditional BNPL loans. Prior work on BNPL studies its effects on access to credit and spending (Akana and Doubinko 2023, Kumar et al. 2024), and finds that BNPL facilitates overborrowing, thereby adversely affecting consumer financial health (deHaan et al. 2024, Guttman-Kenney et al. 2023). On the merchant side, BNPL operates through price discrimination mechanisms, driven primarily by liquidity constraints and differences in product quality (Desai and Jindal 2024). In each of these settings, however, financing is provided by a third-party institution whose incentives are governed solely by credit risk and repayment, and the underlying products are not collateralized.

To summarize, prior research largely treats financing and trading as distinct functions carried out by separate agents. The tokenomics literature views tokens primarily as coordination mechanisms, paying limited attention to their role in facilitating peer-provided, collateralized lending. We address these gaps by examining a market in which participants simultaneously trade and extend credit to one another, a structure made possible by tokenization, which allows assets to serve as both tradable goods and loan collateral. This setting enables us to assess whether embedded financing increases market activity and broadens participation, and whether it creates platform-specific advantages or generates spillovers to competing platforms, offering direct implications for the design of such marketplaces.

3. Institutional Details and Data Description

Starting May 2023, Blur offered financing by introducing a policy referred to as *BNPL*. This policy was introduced in a staggered manner between May 2023 and December 2023 for 13 top-performing collections in the PFP category. These collections accounted for 51% of Blur's sales volume (USD) in May 2023.[9] Although these NFTs can be traded on multiple platforms, the financing option was exclusive to Blur, i.e., an NFT listed on Blur with financing could potentially be purchased on OpenSea, but without financing.

3.1. Institutional Details

3.1.1. Loan Supply: After Blur enabled financing, financing is provided directly by users rather than by the platform. Users who wish to act as lenders make capital available by posting loan offers that can be used by buyers to finance NFT purchases. Each loan offer specifies a set of contract terms, including the maximum loan amount, interest rate (APY), maximum loan-to-value ratio (which determines the fraction of the NFT's purchase price that can be financed), etc. Loan offers are associated with specific NFT collections approved for financing, and a user can avail the loan only if they purchase an NFT belonging to the specific collection.

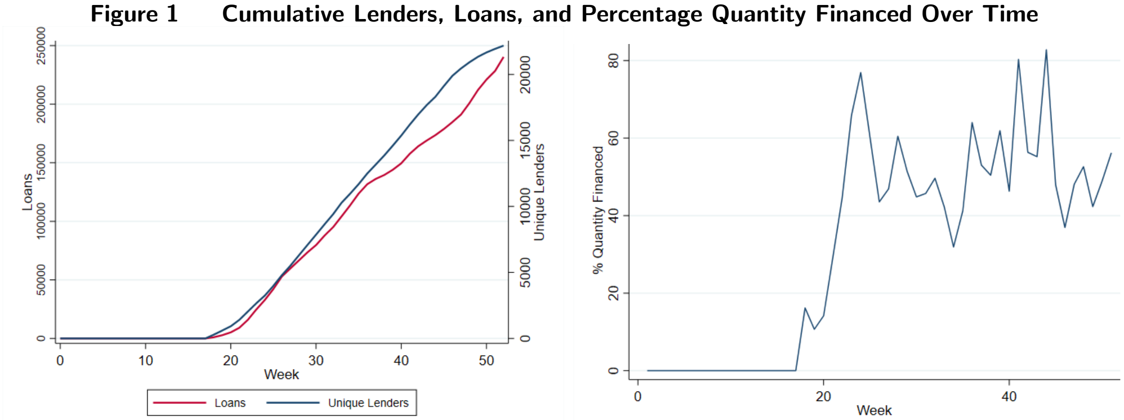
These loan offers are displayed to buyers at the point of purchase.[10] When a buyer selects financing, they choose from the available loan offers instead of paying the full price up front. As a result, lenders compete with one another by offering more attractive terms, such as lower interest rates or higher loan-to-value ratios, in order to have their capital used. This mechanism resembles peer-to-peer lending markets in which loan supply and pricing are determined by individual lenders rather

[9] See `https://blog.xp.network/blend-blurs-new-nft-lending-protocol-all-you-need-to-know-fc4c68730947` for more details. We inferred the introduction of financing options for each NFT collection by tracking their announcements on X (`https://twitter.com/blur\_io`). For each collection, the announcement date coincided with the availability of loans on the platform.

[10] The lower panel of Figure 6 in Appendix shows typical loan offers availed by users purchasing various NFTs in the BAYC collection.

than a centralized intermediary (Vallee and Zeng 2019, Balyuk 2023). This structure differs from traditional BNPL arrangements, where a single financial intermediary sets loan terms and bears the credit risk. In Blur's setting, loan pricing reflects lenders' own assessments of risk, including expected NFT price volatility, liquidity of the underlying asset, and the likelihood of borrower default. Credit risk is mitigated through collateralization, as the financed NFT itself serves as collateral for the loan. Figure 1 shows a consistent increase in the number of loans and unique lenders after financing was enabled (left panel), and a stable 55-60% uptake of financing (right panel). At the aggregate level, thus, we find evidence of both the availability of loans and engagement by lenders, and utilization of loans by buyers.

Figure 1 Cumulative Lenders, Loans, and Percentage Quantity Financed Over Time



Note. Panel (a) plots the cumulative number of loans (red line) and unique lenders (blue line) participating in the financing mechanism on Blur over the 52-week sample period. Panel (b) plots the percentage of quantity financed on Blur over the 52-week sample period.

3.1.2. Incentives for Lending: Lenders commit capital (Ethereum) at the time of lending but accrue interest only after the loan has been availed. To compensate lenders for having their money locked up, Blur implemented a system of lending points. In this system, lenders obtain points only while a loan offer is listed and remains unaccepted, and the points accrued depend on the attractiveness of the loan terms, such as the offered annual percentage yield (APY). Two points merit discussion. First, borrowers do not receive any points for taking out loans, so any increase in purchases or loan utilization is driven by financing rather than lending points. Second, lending points are earned only until the loan is availed, i.e., the lender stops accruing lending points when the loan is availed and begins earning the specified interest on the loan. Lending points are similar to the fees (typically 2% to 8%) merchants pay BNPL providers (in conventional settings) to stimulate financing.[11] During the period of our analysis, Blur provided additional incentives

[11] For more details, please refer to `https://www.shopify.com/in/blog/buy-now-pay-later` and `https://medium.com/@dkschong/earning-lending-points-on-blur-56242e7be908`.

in the form of listing and bidding points to encourage buying and selling. These incentives are available across all collections, not only the ones for which financing is enabled, and are unlikely to confound our main estimates of the effect of financing.[12]

Table 1 Loans, Refinances, and Defaults by Collection in 2023

Collection	# Loans Availed	Loan Uptake	APY	% Refinanced	Refinanced APY	% Repaid	% Seized
Azuki	67,993	66%	22%	66%	11%	36%	0.64%
Milady	56,647	37%	26%	78%	14%	24%	0.78%
Mutant Ape Yacht Club	63,578	62%	27%	58%	16%	43%	0.61%
Pudgy Penguins	35,991	36%	27%	69%	12%	31%	0.39%
Bored Ape Yacht Club	29,776	91%	16%	63%	9%	40%	0.49%
BEANZ Official	25,374	11%	25%	66%	24%	34%	2.35%
CloneX	20,751	14%	21%	75%	12%	25%	0.96%
Bored Ape Kennel Club	20,612	25%	28%	68%	15%	33%	1.05%
Remilio Babies	17,712	11%	32%	77%	33%	26%	1.95%
Otherdeed for Otherside	15,753	3%	22%	82%	23%	20%	1.92%
Kanpai Pandas	8,222	14%	31%	78%	22%	22%	1.33%
Moonbirds	5,991	6%	34%	57%	23%	36%	1.6%
Lil Pudgys	3,892	3%	37%	56%	25%	38%	0.64%

Note. The Table reports lending activity for each of the 13 financing enabled collections on Blur during 2023. Columns report the total number of loans taken, percentage uptake of loans, average annual percentage yield (APY), percentage of refinanced loans, average refinance APY, percentage of repayments, and percentage of collateral seizures. APY reflects the interest rate offered by lenders at the time of loan origination.

3.1.3. Loan Acceptance: Buyers may either pay the full price upfront or select from available financing offers. When a buyer accepts a loan, the NFT serves as collateral. Ownership rights are conditional on repayment, and default is automatically enforced by the smart contract. If the borrower defaults, the lender acquires the NFT, thereby mitigating credit risk through collateralization. The first three columns of Table 1 summarize lending activity across major NFT collections on Blur during 2023, reporting the number of loans availed, the uptake rate of loans, and the average annual percentage yields (APYs). Although there exists variation across collections in terms of loan activity, we consistently find that a substantial proportion of the available loans are actually utilized, and about 23.4% of transactions utilize a loan. Average APYs on initial loans range from approximately 16% to over 35%, reflecting differences in perceived risk and price volatility.

3.1.4. Refinancing: Given the volatile nature of the market, the platform allows lenders the flexibility to recall or end loans at any time. If a lender chooses to end a loan, the platform automatically attempts to refinance the loan through other loan offers that are available for the particular collection. Essentially, refinancing doesn't place the onus directly on the borrower, as the platform automatically attempts to find another lender first. If the platform's effort to refinance is unsuccessful, the borrower can either repay the loan and acquire full ownership of the asset, or

[12] Appendix 7.1 provides more details on this. Additional discussion on other changes in the NFT ecosystem affecting both Blur and OpenSea during our sample period is available in Appendix 7.2.

can default, in which case, the tokenized asset is seized by the lender.[13] Table 1 shows that a large majority of instances where the lender recalls a loan result in the platform automatically refinancing the loan (Column 4) at lower APYs as compared to the original loans (Column 5). The lower APYs likely reflect not only lower uncertainty and better quality NFTs getting refinanced, but also the possibility of active renegotiation of loan terms over time. Despite the high volume of lending activity, default rates, measured by the share of loans resulting in collateral seizure, averaged 1.21%, indicating that the majority of loans are repaid or refinanced rather than seized. Overall, the table illustrates that peer-provided lending is widely used across collections, characterized by refinancing and low realized default rates.

3.2. Data Description

We collect a unique transaction-level dataset spanning 52 weeks from January to December 2023, including trading data for each NFT across 13 treated collections and 36 additional popular collections spanning categories such as PFP, Gaming, Art, and Membership.[14] Together, these collections account for 66% the total trading volume of Blur. Each collection is identified by a unique collection ID, and each NFT within a collection is identified by a unique token ID. We obtain daily data on trading activity across Blur and OpenSea using OpenSea and other third-party APIs. For each platform, we aggregate transactions at the collection–week level to construct measures of weekly quantities sold and average prices (in USD). We exclude wash trades from our sample using the open-source detection filters developed by Dune Analytics, which identify self-trades, back-and-forth trades, repeated purchases of the same token, and trades between commonly funded wallets.[15] This filter-based approach follows established practice in the literature on crypto-asset market manipulation (Cong et al. 2023, Aloosh and Li 2025, von Wachter et al. 2022) and corresponds to the direct detection filters validated in recent work on NFT markets (Hemenway Falk et al. 2025).

In addition, we collect data on new and active users at the collection level. Each user is associated with a unique identifier, allowing us to distinguish between demand responses of existing users and new users. We also compile listing information for NFTs regardless of whether they are ultimately sold. In this setting, an NFT listed on one platform can be sold on another platform, i.e., is interoperable. Therefore, we assume that listing on one platform is equivalent to listing on the

[13] https://www.paradigm.xyz/2023/05/blend

[14] Note that financing was enabled for 15 collections during our sample period. We exclude 2 collections — Wrapped CryptoPunks and DeGods — from our sample. Wrapped CryptoPunks were excluded because the tokens within the collection frequently change due to the wrapping and unwrapping process, making it difficult to track consistent trading activity. DeGods was excluded because the collection migrated from Solana to Ethereum in March and enabled financing in May, leaving an insufficiently short pre-treatment period for meaningful comparison.

[15] See `https://dune.com/blog/nft-wash-trading-on-ethereum`.

other platform as well, i.e., listing is not specific to a platform. Finally, for each transaction, we observe whether financing was used and, when applicable, the lender's identity. Table 2 summarizes the characteristics for both financing enabled (treated) collections as well as collections for which financing was not available (control) at the category level. The detailed collection-level statistics for each of these categories are provided in Appendix Tables A1, A2, A3, and A4.

Table 2 Financing-Enabled Collections and Category-Level Weekly Averages

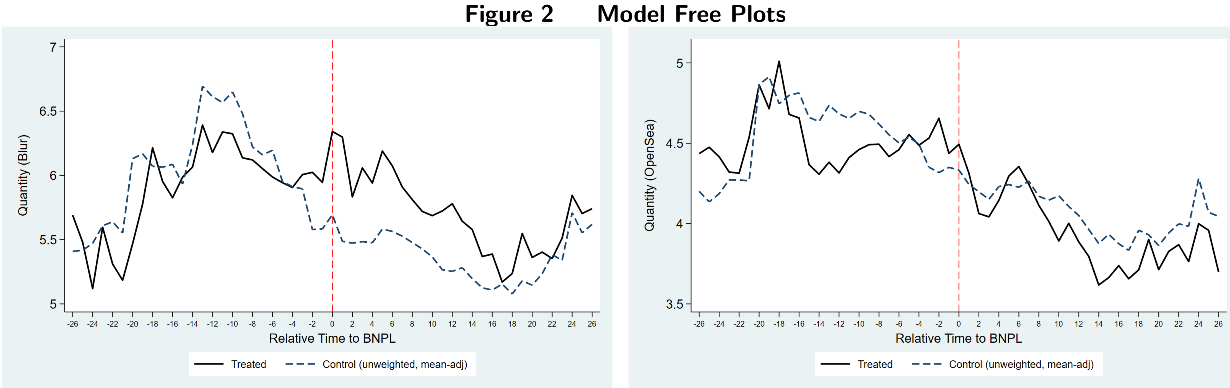
Category	Avg. Qty.	SD Qty.	Avg. Price	SD Price	Collection Size	Users
Collections with Financing	737.20	740.87	12,000.90	21,992.43	15,070.57	11,351.71
Other PFP Collections	399.22	416.98	4,439.39	2,348.26	14,999.00	7,325.50
Art Collections	561.03	1,128.27	1,184.63	2,027.42	13,750.8	9,614.21
Membership Collections	141.55	435.44	2,141.15	5,469.60	7,155.7	2,344.87
Gaming Collections	127.21	153.27	953.34	1,010.29	9,852.7	2,917.42

Note. The Table reports category level weekly averages for financing enabled collections and non-financing-enabled collections across four categories: Other PFP, Art, Membership, and Gaming. Columns report the average weekly units sold, the standard deviation of weekly quantity, the average selling price (in USD), the standard deviation of price, the average collection size (number of unique NFTs), and the average number of active users per week. All statistics are computed over the 52-week sample period from January to December 2023.

Figure 2 shows the natural logarithm of weekly quantities sold of treated and control collections around the introduction of financing, separately for each platform, Blur (left) and OpenSea (right). As financing was introduced at different weeks for different collections (a staggered rollout), we align all cohorts to a common event-time axis by expressing each collection's data in relative time, where Relative Time = 0 denotes the week financing was enabled for that cohort. The control group is the average of control collections for which financing was never enabled (i.e., were never treated), with a mean adjustment to account for time-invariant level differences between treated and control collections.

Prior to the introduction of financing, treated and control collections broadly exhibit similar sales patterns. Following the availability of financing, however, treated collections maintain visibly higher sales levels relative to control collections, but this effect persists only at Blur (left panel) and not at OpenSea (right panel). The overall decline in sales before and after the treatment is consistent with a broader market downturn, as can be seen in Figure 3, which plots total monthly sales of NFTs over the same period. Treated and control collections on OpenSea exhibit similar sales patterns before and after the introduction of financing at Blur. The absence of a post-treatment gap between treated and control collections on OpenSea indicates that the increase in relative sales observed for treated collections is specific to Blur and that the financing impact does not spill over to the rival platform.

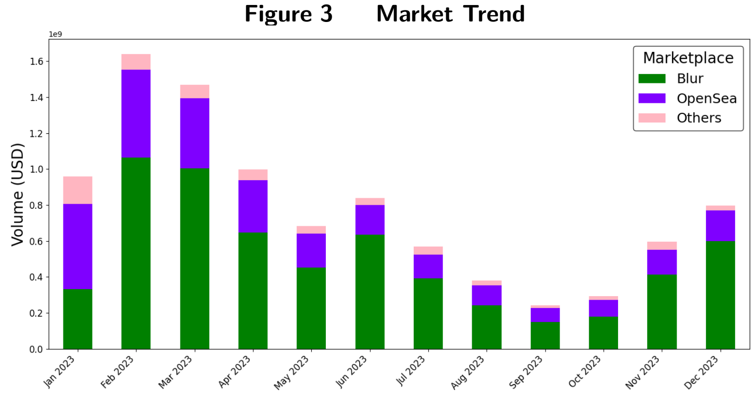
Figure 2 Model Free Plots



(a) Quantities sold at Blur (b) Quantities sold at OpenSea

Note. The Figure plots the average log unit quantity sales for treated and control collections around the introduction of financing. All cohorts are aligned to a common event-time axis, where Relative Time = 0 denotes the week financing was enabled for each cohort. The control group is the average of control collections with a mean adjustment to account for level differences that are constant over time. Panel (a) shows sales on Blur; Panel (b) shows sales on OpenSea.

Figure 3 Market Trend



Note. The Figure plots the monthly trading volume (in USD) across Blur, OpenSea, and Others marketplaces from January to December 2023. The Figure illustrates the broader market downturn observed during the sample period.

Taken together, these model-free patterns suggest that financing is associated with higher quantities sold for treated collections on the platform where financing is offered, while leaving sales on the rival platform largely unaffected. As treated and control collections differ along several dimensions, these results are suggestive and motivate the causal analysis.

4. Empirical Approach

Given the observational nature of our data, there are several specific challenges in identifying the treatment effect of financing. First, the collections in our data are either treated (i.e, financing enabled) or untreated, and once treated, they stay in the treated condition. Therefore, we do not observe counterfactual outcomes for all conditions. Second, Blur enabled financing for leading collections, which are likely to be different from other collections. More broadly, the decision to enable financing for a collection may be driven by unobservable collection characteristics that may also be correlated with the performance of that collection. Third, the timing of enabling financing varies across collections and could be endogenous. Additionally, the staggered timing could potentially result in dynamic treatment effects that require careful aggregation and computation of the estimates. Finally, given that the platform's decision to enable financing was made at the collection level, differences in user substitution between treated and non-treated collections may be related to financing. Such substitution would violate the Stable Unit Treatment Value Assumption (SUTVA) (Goldfarb et al. 2022), the validity of which is critical for identification by comparing treated and control groups.

To address the first challenge, we create a carefully selected control group of collections. Ideally, to make an appropriate comparison, we not only need the control group to be similar to the treatment group, but also need to have no substitution across control and treatment groups. To balance and trade-off these competing factors, we include in the control group two other popular NFT collections from the PFP category, and top-performing collections from other popular categories such as Art (11 collections), Gaming (10 collections), and Membership (13 collections), which are as popular as PFP but are unlikely to be substitutes for the PFP category. Ho et al. (2024) provide empirical support for this approach by demonstrating that cross-category substitution in NFTs is low. At a high level, the control group includes collections that are similar to treated collections either through category or popularity, yet are less likely to be substitutes for the treated collections.

We use the TWFE model to estimate the impact of financing on user behavior, where we regress the natural logarithm of the outcome Y_{it} of the collection i in week t on the interaction between the treatment indicator $Treatment_i$ and the post-treatment period indicator variable $Post_{it}$ as follows:

$$\ln(Y_{it} + 1) = \beta_0 + \beta_1 Treatment_i \times Post_{it} + \gamma_i + \eta_t + \epsilon_{it} \tag{1}$$

where collection (γ_i) and period (η_t) fixed effects account for collection-specific time-invariant and aggregate-level time-varying differences, respectively (Wooldridge 2021). The collection fixed effects absorb any time-invariant differences between collections, such as collection size or creator reputation. The time fixed effects absorb market-wide shocks that affect all collections equally in a given week, such as broader market sentiment. The coefficient β_1 then captures the average change in the outcome for treated collections after financing is enabled, relative to control collections, after accounting for these fixed differences.

Although the collection fixed effects partially address our second challenge, the estimates may still be biased if the parallel trends assumption does not hold or if the linearly additive effects assumption is invalid (Imai and Kim 2021). We test the parallel trends assumption using a relative-time specification and find no evidence of systematic pre-treatment differences between treated and control collections, providing support for the validity of the assumption (see Section 5.4). However, concerns remain due to the staggered adoption of financing. In such settings, the standard two-way fixed effects DiD estimator can produce biased estimates because it aggregates multiple two-group, two-period comparisons and may assign negative weights to some comparisons when treatment effects are heterogeneous across cohorts or over time (Goodman-Bacon 2021). While the Two-Stage DiD estimator (Gardner 2022) addresses these issues associated with staggered treatment timing, it does not explicitly account for dynamic treatment effects and continues to rely on the parallel trends assumption.

To address these concerns, we use the Synthetic DiD estimator (Arkhangelsky et al. 2021) as our primary empirical specification. Synthetic DiD constructs a weighted synthetic control for each treated collection by assigning unit and time weights to observations in the control group. These weights are chosen so that the synthetic control closely matches the pre-treatment trajectory of the treated collection, thereby improving comparability between treated and control units. The resulting synthetic control provides a more credible counterfactual against which to evaluate the effects of financing. To accommodate staggered treatment adoption, we apply the Synthetic DiD estimator separately to groups of collections sharing the same financing adoption date and then aggregate the resulting estimates (Arkhangelsky et al. 2021).

A related concern is that financing may have been introduced for collections experiencing favorable demand shocks that are difficult to observe. For example, increases in collection popularity, media attention, creator activity, or broader demand for a collection may simultaneously affect both the likelihood of financing adoption and marketplace outcomes. Although Synthetic DiD improves comparability between treated and control collections by matching pre-treatment trends, it may not fully eliminate concerns regarding collection-specific shocks that evolve over time. Importantly, an important feature of our setting is that the same NFT collections are traded simultaneously on multiple marketplaces. Because financing was introduced only on Blur, while the same collections continued to trade on OpenSea without financing, we can additionally compare outcomes for the same collection across marketplaces. This allows us to estimate a triple-difference specification that differences out collection-level shocks common to both platforms and provides an additional source of identification.

Finally, there is a potential violation of SUTVA if users substitute between the treated and control group collections. Since our control group predominantly includes collections from different categories, substitution across collections is less likely a concern (Ho et al. 2024). Having said this, we conduct robustness checks either by excluding overlapping users between treatment and control collections or by restricting control group collections to those with minimal overlap with treated collections, and find no qualitative differences in the estimates (results reported in Section 5.4). We outline the Two-Stage DiD, Synthetic DiD, and Triple Difference Design specifications below.

4.1. Two-Stage DiD

Two-Stage DiD (Gardner 2022, Arkhangelsky et al. 2021) enables us to address the challenge due to staggered adoption, if any, as it involves computing the treatment effects for each cohort separately and finally aggregating them to get an average effect on the treated. Moreover, it addresses the negative weighting of treatment effects, which may possibly lead to biased estimates (Goodman-Bacon 2021).

To estimate our Two-Stage DiD model, we follow the approach proposed by Gardner (2022), which involves multiple collections treated at different points in time. The central concern with staggered adoption is that treatment effects may be heterogeneous across groups and over time. For example, initial collections for which financing is enabled may respond differently compared to later ones, resulting in heterogeneous treatment effects across groups and periods. Such challenges are addressed in the approach proposed by Gardner (2022), by restricting the first-stage estimation of fixed effects to only untreated and not-yet-treated observations. In a standard TWFE regression, fixed effects are estimated using the entire sample, including post-treatment observations from already-treated units, which means the treatment effect contaminates the estimated fixed effects. When treatment effects are heterogeneous, this contamination causes the estimated counterfactual to be biased. By excluding the treated observations from the first stage, the approach in Gardner (2022) ensures that the counterfactual is not contaminated. The treatment effect is then estimated in a second stage on the full sample. Moreover, this approach retains the efficiency advantages pointed out by Borusyak et al. (2024). The estimation follows a two-step process. In the first stage, the approach makes use of the subsample of untreated/not-yet-treated observations, retaining the estimated group and time effects to form the adjusted outcome variable. The first stage is estimated as follows:

$$Y_{igt} = \mu_g + \eta_t + \epsilon_{igt} \tag{2}$$

where Y_{igt} is the outcome variable of interest (collection's sales, etc.), i represents the individual collections, t denotes time period, and g indicates group membership where a group is represented by all collections whose treatment begins at time g. μ_g represents time-invariant group characteristics, and η_t represents the unobserved shocks in a given time period that impact all collections. The adjusted outcomes $\widehat{Y_{igt}}$ are then regressed on the treatment status in the full sample to estimate treatment effects as follows:[16]

$$\widehat{Y_{it}} = \beta_0 + \beta_1 Treatment_i \times Post_{it} + \epsilon_{it} \tag{3}$$

As the fixed effects are estimated exclusively from untreated observations, the adjusted outcomes for treated collections are not contaminated by other cohorts' treatment effects. This ensures that

[16] This is calculated as $\widehat{Y}_{igt} = Y_{igt} - \widehat{\mu}_g - \widehat{\eta}_t$.

heterogeneous effects across collections or over time do not bias the overall estimate, i.e., each treated collection's residual reflects only its own treatment effect, and the second-stage regression averages these with equal-share weights rather than the potentially negative weights that arise under TWFE (Goodman-Bacon 2021).

4.2. Synthetic DiD

In order to implement Synthetic DiD, we create a balanced panel, which is an essential requirement for the algorithm. Following this, we calculate the weights $\widehat{\omega}$ based on the pre-treatment trends of the performance variable for both the treated and control units. Let $N_{control}$ represent the collections in the control group, and N_{tr} stand for the collections in the treated group, and $T = 1, \ldots, T_{pre}$ represent the pre-treatment period. The unit weights are computed as follows:

$$(\widehat{\omega_0}, \widehat{\omega_i}) = \underset{\omega_0 \in \mathbf{R}, \omega_i \in \Omega}{\arg\min} \sum_{t=1}^{T_{pre}} \left(\omega_0 + \sum_{i \in N_{control}} \omega_i Y_{it} - \frac{1}{N_{tr}} \sum_{i \in N_{tr}} Y_{it} \right)^2 + \zeta T_{pre} ||\omega||_2^2 \tag{4}$$

where $\widehat{\omega_0}$ denotes the intercept and $\widehat{\omega_i}$ are the weights assigned to the collection i. Ω denotes the unit simplex, i.e., the set of non-negative weights that sum to one. ζ stands for the regularization parameter to lower the variance and to ensure the uniqueness of the weights. Y_{it} stands for our dependent variable of interest, i.e., the performance of the NFT collections measured by weekly sales. In addition to assigning weights to the collections to ensure that the treatment units are comparable to the control units, the method also ensures a constant difference between the average post-treatment outcomes and weighted pre-treatment outcomes of the control unit by incorporating time weights λ_t. The time weights are created using the pre-treatment time periods similar to the treatment periods for the control group, as follows:

$$(\widehat{\lambda_0}\ ,\ \widehat{\lambda}) = \underset{\lambda_0 \in \mathbf{R}\ \ \lambda_t \in \Lambda}{\arg\min} \sum_{i \in N_{control}} \left(\lambda_0 + \sum_{t \in T_{pre}} \lambda_t Y_{it} - \frac{1}{T_{post}} \sum_{t \in T_{post}} Y_{it} \right)^2 \tag{5}$$

where $\widehat{\lambda_0}$ denotes the intercept and $\widehat{\lambda_t}$ are the weights for time t. Using the collection and time weights, the estimates for Synthetic DiD are then obtained by optimizing the following:

$$(\widehat{\phi},\ \widehat{\mu},\ \widehat{\alpha},\ \widehat{\beta}) = \underset{\phi, \mu, \alpha, \beta}{\arg\min} \left(\sum_{i=1}^{N} \sum_{t=1}^{T} (Y_{it} - \mu - \phi Treatment_{it} - \alpha_i - \beta_t)^2 \, \widehat{\omega}_i \widehat{\lambda}_t \right) \tag{6}$$

where ϕ represents the average treatment effect of financing on the performance of the NFT collections. α_i and β_t represents collection and time fixed effects, respectively. The unit and time weights are represented by $\widehat{\omega}_i$ and $\widehat{\lambda}_t$, respectively. $\widehat{\omega}_i$, created by matching and reweighting the collections in the control group, ensures that these units are similar to the treated units. $\widehat{\lambda}_t$ created by matching and reweighting in the time dimension ensures that the pre-treatment period is similar to that of treatment times. This makes the identifying assumption substantially more credible when treated and control units differ on observables, though it does not eliminate the need for an assumption about counterfactual outcomes (Arkhangelsky et al. 2021). We assess the credibility of this assumption through pre-trend visualizations and the robustness checks reported in Section 5.4.

In the presence of staggered adoption, the above optimization (Equations 4–6) is performed separately for each treatment cohort, defined as the set of collections for which financing is enabled in the same week Arkhangelsky et al. (2021). Each cohort yields its own set of unit weights, time weights, and treatment effect estimate. The overall average treatment effect on the treated is then obtained by calculating a weighted average of the cohort-specific estimates, where weights are assigned based on the relative number of treated units and time periods in each adoption group.

4.3. Triple Difference Design

An important feature of the NFT market is that the same collection is frequently traded simultaneously across multiple marketplaces. While financing was introduced on Blur, the same NFT collections continued to trade on OpenSea without access to financing. This setting provides an additional source of identification that allows us to compare outcomes for identical assets across treated and untreated marketplace environments.

Specifically, we exploit variation across collections, time, and marketplaces to estimate a triple-difference (DDD) specification. The DDD design compares changes in outcomes for treated collections relative to control collections before and after financing adoption, while additionally differencing these changes across Blur and OpenSea. The identifying intuition is that any collection-specific shock that affects demand for a collection, such as changes in popularity, media attention, creator activity, or broader collection-level trends, should influence trading activity on both marketplaces. By comparing the same collection across platforms, the DDD estimator differences out such collection-level shocks and isolates the effect of financing on Blur.

Formally, we estimate:

$$Y_{ipt} = \beta\,(Treated_i \times Post_{it} \times Blur_p) + \gamma_{ip} + \delta_{it} + \eta_{pt} + \epsilon_{ipt} \tag{7}$$

where Y_{ipt} denotes the outcome for collection i on platform p in week t. $Blur_p$ is an indicator equal to one for observations from Blur and zero for OpenSea. The specification includes collection-platform fixed effects (γ_{ip}), collection-week fixed effects (δ_{it}), and platform-week fixed effects (η_{pt}).

The coefficient β captures the effect of financing on Blur relative to OpenSea. The collection-week fixed effects absorb all collection-specific shocks that are common across platforms in a given week, while the platform-week fixed effects absorb marketplace-wide shocks affecting all collections on a platform. As a result, identification comes from the differential change in activity on Blur relative to OpenSea following the introduction of financing.

The validity of this approach relies on the assumption that, absent financing, treated and control collections would have experienced similar changes in the Blur–OpenSea outcome differential over time. The DDD specification complements our Synthetic DiD approach. Whereas Synthetic DiD constructs a weighted counterfactual using untreated collections, the DDD design uses the same collection trading on a competing marketplace as an additional counterfactual.

While the DDD design provides further control for collection-level confounding factors, it relies on its own identifying assumptions. Specifically, in the absence of financing, treated and control collections must exhibit similar evolution in the Blur–OpenSea outcome differential over time. In addition, the design assumes that the introduction of financing on Blur does not generate substantial spillovers that affect activity on OpenSea. We test these assumptions by directly examining outcomes on OpenSea and estimating dynamic treatment effects. The evidence reveals little indication of cross-platform spillovers (shown in Section 5) or differential pre-treatment trends, lending support to the validity of the DDD design.

5. Results

Table 3 reports the effect of enabling financing (treatment) for a collection of NFTs on the quantity and prices at Blur (Columns 1-2) and OpenSea (Columns 4-5) using different approaches. In Column 3, we report the impact of financing on the total number of listings, which, as we discuss later, is a supply-side measure. The top panel reports these estimates based on all transactions, whereas the bottom panel reports estimates for Blur based only on transactions that did not use financing.

Impact of Financing on Blur: Focusing on the estimates from Synthetic DiD (our main specification) in the top panel, we find that offering financing at Blur on average leads to a 49% ($= e^{0.398} - 1$) increase in quantity of NFTs transacted on the platform (Model 1). Financing does not have a statistically significant effect on the average selling price at Blur (Model 2) which, combined with the increase in quantity sold, points to the ability of financing to increase transactions without having any distortionary effect on prices. These results are qualitatively (and quantitatively) robust to different model specifications (TWFE and Two-Stage DiD in rows 2 and 3, respectively).

The estimates above capture the effect of the availability of financing on marketplace outcomes. To the extent that Blur advertised enabling financing for these collections, it is possible that the observed increase in transactions could stem from advertising effects as opposed to the actual utilization of loans. We test this in two different ways. First, we test whether the increase in quantity sold after financing was enabled is driven by the actual use of financing rather than

Table 3 Impact of Financing on Collection Outcomes

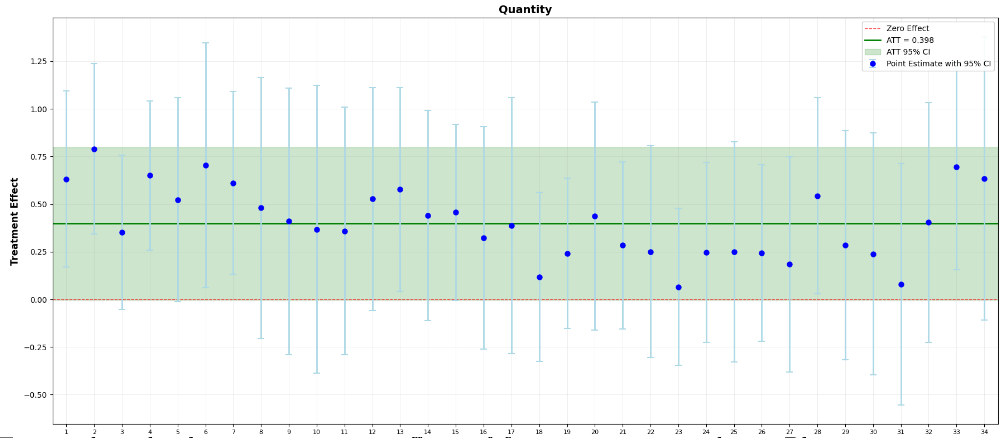
	Blur			OpenSea	
	Quantity (1)	Price (2)	# Listings (3)	Quantity (4)	Price (5)
All Transactions					
Synthetic DiD	0.398**	-0.097	0.983***	0.036	-0.030
	(0.187)	(0.163)	(0.436)	(0.167)	(0.152)
Two-Stage DiD	0.522**	0.206	0.757	0.045	0.225
	(0.218)	(0.220)	(0.480)	(0.137)	(0.219)
TWFE	0.528**	0.117	0.822**	0.067	0.182
	(0.249)	(0.234)	(0.482)	(0.159)	(0.408)
Non-Financed Transactions					
Synthetic DiD	0.210	-0.098	–	–	–
	(0.200)	(0.186)			

Note. The top panel of the Table reports estimates from different models (in rows) with quantity, prices, and number of listings based on all transactions at Blur and OpenSea as the dependent variables (in columns). The bottom panel of the Table reports the same estimates using the Synthetic DiD estimator, but based only on non-financed transactions, i.e., transactions that did not use financing in the pre- and post-treatment periods. All models are based on 2,499 collection-week level observations and include collection and week fixed effects. Standard errors reported in parentheses are clustered at the collection level. Asterisk indicate significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

its mere availability. To do this, we explore the change in the number of transactions which do not use financing before and after financing is introduced. If users increase purchases but do not utilize financing, then we should see an increase in non-financed transactions similar to that in the top panel. If, however, the increase in transactions is driven by financing, then we should see no significant change in the number of transactions without financing. Thus, we exclude all transactions executed using financing on Blur and re-estimate our model (bottom panel of Table 3). We find no statistically significant effect of financing on transaction volume in this specification, which points to the baseline results being driven by users' active use of financing rather than by the mere presence of financing. We also examine how the effect evolves over event time. Figure 4 plots the dynamic effect of financing on sales, with vertical bars showing the 95% confidence intervals. The effect is positive in the post-treatment period. The higher standard errors in the later weeks are driven partly by less data in these weeks, given the staggered nature of treatment.

Impact of Financing on OpenSea: Enabling financing is a strategic decision by a platform the effect of which can potentially spillover from the focal platform to other platforms. The fact that NFTs are interoperable makes this more likely. We examine whether there exist any spillovers from financing on competing platforms, which also provides another test of whether the increase in transactions could be driven by collection-specific unobservables. Columns 4-5 of Table 3 report the impact of financing on OpenSea, the prominent competing platform. We do not find any impact of

Figure 4 Impact of Financing on Blur over Time



Note. The Figure plots the dynamic treatment effects of financing on unit sales at Blur over time, estimated using the Synthetic DiD model. Each point represents the period-specific treatment effect estimate, and the vertical bars indicate the 95% confidence interval. The horizontal green line denotes the average treatment effect on the treated (ATT), and the dashed line indicates the zero-effect benchmark.

financing at Blur on the quantity (Model 4) of the same collection at OpenSea. Thus, there exist no direct spillovers (positive or negative) of financing on the sales at rival platform. Additionally, we do not find any effect of Blur's financing on the prices of NFTs at OpenSea.[17]

Triple Difference Estimator: We complement these estimates with a triple-difference (DDD) specification that exploits the fact that the same collections trade simultaneously on Blur and OpenSea, while financing was introduced only on Blur. Any collection-specific demand shock should affect both platforms, whereas financing affects only Blur. By differencing outcomes across platforms, the DDD estimator absorbs collection-specific shocks common to both marketplaces and isolates the incremental effect of financing on Blur relative to OpenSea. The validity of this design relies on the absence of spillovers to the comparison platform, since financing on Blur should not affect activity on OpenSea. We find no effect of financing on quantities or prices at OpenSea (Columns 4–5 of Table 3), which supports this assumption. Table 4 reports the results. Consistent with our main findings, financing has a positive and statistically significant effect on quantities sold, while having no significant effect on prices (top panel). When we restrict the sample to non-financed transactions (bottom panel), the effect on quantities falls in magnitude and is no longer statistically significant, mirroring the non-financed results in Table 3. This pattern indicates that the increase in activity is driven by the actual use of financing rather than by its mere availability.Taken together, these findings strengthen our identification strategy by demonstrating that the observed effects are specific to the marketplace where financing is introduced rather than reflecting collection-level demand shocks or broader market trends that influence activity on both platforms simultaneously.

[17] In Appendix, Figure 7, we explore whether the aggregate null effect masks any dynamics in spillover at OpenSea over time. We consistently estimate a null effect of financing on OpenSea in all subsequent periods.

Table 4 Triple Difference Estimator

	Quantity (1)	Price (2)
All Transactions		
Post × Treated × Blur	0.453**	-0.119
	(0.183)	(0.074)
Non-Financed Transactions		
Post × Treated × Blur	0.274	0.172
	(0.175)	(0.094)

Note. The Table reports estimates from triple-difference (DDD) specifications that compare outcomes across treated and control collections, before and after financing adoption, and across Blur and OpenSea. The coefficient on *DDD* captures the effect of financing on Blur relative to OpenSea after accounting for collection-level and time-varying factors common to both marketplaces. The top panel uses all transactions; the bottom panel uses only transactions that did not use financing in the pre- and post-treatment periods. All models are based on 4,998 collection-week level observations and include the fixed effects implied by the DDD design. Standard errors reported in parentheses are clustered at the collection level. Asterisk indicate significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Our analysis so far explores the impact of financing on quantities and prices, but does not speak to the supply side response. Availability of financing is likely to attract a response from creators and sellers. In this setting, the number of NFTs in a collection are pre-written into the blockchain and cannot be modified. Therefore, on the supply side, we could potentially see sellers increasing the number of listings of NFTs for which financing is available. The long-term impact of financing on activity on the platform depends on the extent to which there is an increase in the number of listings as well as how persistent is this effect over time.[18] Model 3 in Table 3 reports the impact of financing on the number of NFTs listed for sale in any week at a collection level. We find that financing results in a 167% ($= e^{0.983} - 1$) increase in the number of listings, i.e., financing increases the supply of NFTs available for purchase. Note that NFTs are typically cross-listed across marketplaces, the same listing appears on both Blur and OpenSea, so it is challenging to segregate listings by platform.

Together, these estimates show that Blur's introduction of financing increases transactions for these collections without distorting NFT prices. The absence of a price increase indicates that supply expanded alongside demand, and the rise in the number of listings confirms this supply response.

[18] Availability of financing on the entire platform could also lead to an increase in the number of collections which are minted. Our analysis does not account for sales stemming from an increase in the number of collections, and thus, the results should be treated as a lower bound on the true supply side response.

5.1. Acquiring New Users or Deepening Relationships?

While the observed increase in transactions following the introduction of financing suggests that financing can stimulate marketplace activity, identifying the underlying source of this growth is critical for assessing its strategic value. Financing may generate sustainable growth if it attracts new users who have not previously participated in the collection. For this, we conduct two analyses. First, we test whether the number of new users purchasing in the collection increases after financing becomes available. Second, we separately study the impact of financing on the quantities purchased by existing users (those who have previously traded in the collection) and new users (users who have not previously traded in the collection).[19]

Table 5 reports the estimates from these analyses for both Blur and OpenSea using different approaches. For both platforms, we do not find any significant effect of financing on either the number of new users purchasing in the category or the quantities sold to new users (rows 1 and 2). By contrast, the number of existing users who make purchases in any week at Blur increases by 75% $(= e^{0.559} - 1)$, and the purchases made by these users increase by 164% $(= e^{0.972} - 1)$. Consistent with aggregate level results, we do not find any increase in the number of existing users or the sales they generate at OpenSea. These results are robust to different estimation methods (Columns 3-6), and provide further evidence that the benefit of financing could be short-lived because of its inability to attract new users and its appeal being concentrated only among users already engaging with the collection.

Taken together, the results indicate that financing promotes growth primarily through the intensive margin by increasing engagement among users who are already familiar with the collection. Although financing does not appear to attract new buyers or sellers in the short run, we find that approximately 30% of lenders are new to Blur and had not previously participated on the platform as either buyers or sellers. Thus, financing does expand participation on the lending side of the marketplace by attracting new capital providers, even if this influx of new users does not immediately translate into additional buying or selling activity.

5.2. Changes in Product Composition

Next, we examine whether the impact of financing varies across high- and low-end NFTs within a collection. Understanding these compositional effects is important because the platform benefits differently depending on which assets experience increased demand. Financing may be less valuable if it primarily increases sales of lower-end NFTs at the expense of higher-end assets. In contrast, if financing relaxes liquidity constraints and enables users to acquire higher-end NFTs that would otherwise be unaffordable, it may increase both transaction volume and marketplace value.

[19] In any period, new and existing users are defined based on all the previous activity of the user within the collection.

Table 5 Impact of Financing on Sales Among Different Users

	Synthetic DiD		Two-Stage DiD		TWFE	
	Blur (1)	OpenSea (2)	Blur (3)	OpenSea (4)	Blur (5)	OpenSea (6)
# New Users making a purchase	-0.006	-0.057	0.213	0.085	0.208	0.106
	(0.153)	(0.144)	(0.163)	(0.123)	(0.182)	(0.132)
Quantity Bought by New Users	0.067	-0.022	0.227	0.047	0.223	0.070
	(0.204)	(0.213)	(0.199)	(0.138)	(0.225)	(0.150)
# Existing Users making a purchase	0.559***	-0.105	0.548***	0.059	0.565***	0.086
	(0.213)	(0.115)	(0.164)	(0.114)	(0.144)	(0.122)
Quantity Bought by Existing Users	0.972***	0.130	0.604***	0.060	0.616**	0.091
	(0.212)	(0.099)	(0.225)	(0.144)	(0.245)	(0.170)

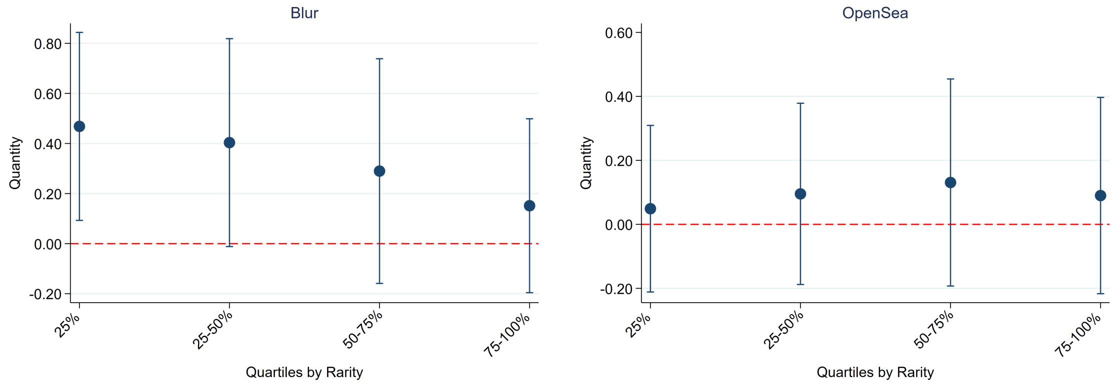
Note. The Table reports the impact of financing on the number of users and sales among new and existing users (in rows) at Blur and OpenSea (in columns). The Tables reports estimates based on Synthetic DiD (Columns 1 and 2), Two-Stage DiD (Columns 3 and 4), and TWFE (Columns 5 and 6). All estimates are based on 2,499 collection-week level observations and include collection and week fixed effects. Standard errors reported in parentheses are clustered at the collection level. Asterisk indicate significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To examine this possibility, we classify NFTs within each treatment and control collection into quartiles based on their rarity scores. Within a collection, NFTs that are unique, scarce, or possess culturally significant attributes tend to have higher rarity scores and are generally perceived as higher-quality assets (Mekacher et al. 2022, Dobrynskaya and Bianchi 2023).[20] Consistent with this interpretation, NFTs in higher rarity quartiles also command substantially higher prices on average. We then estimate the effects of financing separately for NFTs in each rarity quartile.

Figure 5 plots the quartile-specific estimates (from the Synthetic DiD model) of financing on transactions at both Blur (left panel) and OpenSea (right panel). We find a positive and significant effect of financing on the transactions of high-quality NFTs (top two quartiles) at Blur but do not find any effect of financing on the sales of lower-quality NFTs (bottom quartiles). The increase in sales of high-quality NFTs, therefore, is not a result of substitution away from low-quality NFTs. Consistent with the aggregate level results, the impact of financing on transactions of both high- and low-quality NFTs at the rival OpenSea platform is not statistically significant. As is evident from Table 6, we find qualitatively similar results from the Two-Stage DiD and TWFE models as well. These estimates imply an increase of up to 60% ($= e^{0.468} - 1$) in the sales of higher quality NFTs as compared to those in the bottom quartiles. Finally, we also find that financing leads to an increase in the number of listings for both low- and high-quality NFTs, but the increase in listings of low-quality NFTs does not translate to an increase in their sales.

[20] See `https://www.coinbase.com/en-in/learn/crypto-basics/what-is-an-nft-rarity-ranking` and `https://www.lcx.com/what-is-nft-rarity/` for additional discussion of NFT rarity.

Figure 5 Impact of Financing on NFT Quantity by Rarity-based Quartiles on Blur (left) and OpenSea (right)



Note. The Figure plots, for each quartile, the estimate of the impact of financing on the sales of NFTs in that quartile for Blur (left panel) and OpenSea (right panel), respectively. Quartiles are created based on rarity scores with the left most quartile (25%) being the top quartile with NFTs with highest rarity score. The vertical lines indicate the 95% confidence region around the estimate.

Table 6 Financing and NFT Quality

	Synthetic DiD		Two-Stage DiD		TWFE		
Quartile	Blur (1)	OpenSea (2)	Blur (3)	OpenSea (4)	Blur (5)	OpenSea (6)	# Obs. (7)
Top 25%	0.468**	0.049	0.521***	0.140	0.525***	0.158	2,397
	(0.192)	(0.150)	(0.175)	(0.133)	(0.178)	(0.125)	
25-50%	0.404*	0.095	0.493**	0.046	0.491**	0.070	2,040
	(0.244)	(0.116)	(0.209)	(0.115)	(0.247)	(0.129)	
50-75%	0.290	0.131	0.386*	-0.014	0.382	0.005	1,989
	(0.226)	(0.137)	(0.223)	(0.127)	(0.248)	(0.117)	
75-100%	0.151	0.090	0.264	-0.050	0.252	-0.036	1,938
	(0.208)	(0.183)	(0.231)	(0.136)	(0.254)	(0.141)	

Note. The Table reports the impact of financing on quantities sold at Blur and OpenSea (in columns) by quartiles of NFTs (in rows) where within-collection quartiles are defined based on the rarity score. The Tables reports estimates based on Synthetic DiD (Columns 1 and 2), Two-Stage DiD (Columns 3 and 4), and TWFE (Columns 5 and 6). Column 7 reports the number of observations in the analysis for each quartile. All estimates include collection and week fixed effects. Standard errors reported in parentheses are clustered at the collection level. Asterisk indicate significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3. Characterizing Demand- and Supply-Side Responses

Our results show that at an aggregate level, financing leads to an increase in the quantity of the treated collections. From a platform's perspective, it is important to understand the extent to which changes in sales due to financing are driven by supply-versus-demand-side responses. This is especially important if the platform wants to understand the impact of any policy that asymmetrically impacts the supply and the demand side. For example, let's consider a policy where in the presence of financing, the platform wants to impose a tax on lending or charge a fee for listing which directly impacts lenders and sellers, respectively, but does not directly impact the buyers. The profitability of such a policy would critically hinge on how the supply side (sellers) respond to the introduction of financing. The fact that we observe a lack of significant change in prices (Table 3) – despite changing quantities – suggests the presence of both supply- and demand-side responses. We now quantify the relative contributions of demand- and supply-side responses to the total increase in quantities sold.

$$Y_{it}^1 - Y_{igt}^0 = nlist_{it}^1 \pi_{it}^1 - nlist_{it}^0 \pi_{it}^0 = \underbrace{\left(nlist_{it}^1 - nlist_{it}^0\right) \pi_{it}^1}_{supply} + \underbrace{nlist_{it}^0 \left(\pi_{it}^1 - \pi_{it}^0\right)}_{demand} \tag{8}$$

Equation 8 shows that the total increase in sales can be expressed as the sum of two different effects: an increase in the number of listings ($nlist_{it}$) holding fixed the purchase likelihood (π_{it}), i.e. the supply side response, and an increase in purchase likelihood (conditional on a listing) holding fixed the number of listings, i.e. the demand side response. The quantification of relative contributions of supply and demand side is only possible because we observe the listings separately from transactions. We compute the total increase in transactions ($Y_{it}^1 - Y_{igt}^0$) as $(e^{0.398} - 1) * 768.5$ where the estimate 0.398 comes from Table 3 and 768.5 is the average quantity sold per week pre-treatment. Similarly, we compute the supply-side response ($(nlist_{it}^1 - nlist_{it}^0)\,\pi_{it}^1$) as $(e^{0.983} - 1) * 763.2 * 0.115$ where the estimate 0.983 comes from Table 3, 763.2 is the average number of listings per week in the pre-treatment period, and 0.115 is the probability of sale. Based on these numbers, we find that approximately 39% of the total increase in transactions can be attributed to the supply-side change in the number of listings, with the demand-side contributing the remaining 61%. Thus, although the majority of the increase in transactions is driven by an increase in demand, the platform cannot ignore the contribution of supply-side changes. Any policy which asymmetrically influences sellers should therefore, carefully account for the potential supply-side response.

5.4. Robustness Checks

We, first, examine whether treated and control collections exhibit similar trends prior to the introduction of financing. Using a relative-time specification, we find that the pre-treatment leads are statistically insignificant in all but one or two periods for both Blur and OpenSea, with no evidence of systematic pre-treatment differences. Furthermore, joint tests fail to reject the null that pre-treatment coefficients are jointly equal to zero (Blur: $\chi^2(20) = 19.18$, $p = 0.51$; OpenSea: $\chi^2(20) = 20.21$, $p = 0.44$), providing support for the parallel trends assumption. We then conduct a series of robustness checks assessing sensitivity to control-group selection, alternative estimators, potential spillovers between treated and control collections, treatment timing, and the integrity of the underlying transactions.

5.4.1. Alternate Control Group: Second, we examine the sensitivity of our findings to the choice of control group. Our baseline specification uses collections from the art, gaming, and membership categories because they are sufficiently similar to the treated collections while limiting concerns about direct substitution. Nevertheless, it is possible that untreated profile-picture (PFP) collections constitute a more appropriate comparison group because they share similar characteristics and target audiences with the treated collections. To address this concern, we re-estimate our models using the top 75 untreated PFP collections as the control group. These collections, together with the treated collections, collectively account for about 80% of total platform transactions value.

Table 7 reports the results from this analysis for both Blur and OpenSea. Consistent with our baseline findings, financing has a positive and statistically significant effect on quantities sold on Blur, but no significant effect on activity on OpenSea. We also find no significant impact of financing on NFT prices on either platform (Specifications 2 and 4). These results remain qualitatively similar when we estimate the effects using Two-Stage DiD and TWFE models instead of Synthetic DiD. Together, these analyses suggest that our findings are not sensitive to the choice of control group.

Table 7 Robustness Checks with Top 75 PFP Collections as Control Group

	Blur		OpenSea	
	Quantity	Price	Quantity	Price
	(1)	(2)	(3)	(4)
Synthetic DiD	0.413**	-0.133	-0.187	0.022
	(0.175)	(0.189)	(0.174)	(0.147)
Staggered DiD	0.637***	0.144	0.221	0.228
	(0.219)	(0.215)	(0.160)	(0.211)
TWFE	0.651***	0.160	0.241	0.131
	(0.245)	(0.230)	(0.207)	(0.207)

Note. The Table reports estimates for quantity and prices at Blur and OpenSea. All models are based on 4,437 collection-week level observations and include collection and week fixed effects. Standard errors reported in parentheses are clustered at the collection level. Asterisk indicate significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4.2. SUTVA: Third, we address concerns regarding potential spillovers between treated and control collections. Because some users transact across multiple collections, increases in activity in treated collections could partly reflect substitution away from control collections, violating the Stable Unit Treatment Value Assumption (SUTVA). Table 8 shows that user overlap between treated collections and control-group categories ranges from 9.6% to 26.2%. To assess whether such overlap affects our conclusions, we conduct two additional analyses. In the first, we eliminate all users who appear in both treatment and control groups during the pre-treatment period, leaving a sample of 101,620 non-overlapping users. In the second, we restrict the control group to collections with no more than 15% user overlap with treated collections.

Table 8 **Users Overlap Between Treated Collections and Other Categories in the Pre-financing Period**

Category	# Users (1)	Overlapping Users (2)	% Overlapping (3)
Gaming	15,409	6,624	13.20%
Membership	18,118	4,820	9.60%
Art	42,368	13,089	26.20%
Meebits–Doodles	9,917	5,817	11.64%

Note. The Table reports for each category of NFTs (in rows), the number of users making transactions in the category (Column 1), the number (Column 2), and the percentage (Column 3) of users that overlap between the category and the focal 15 treated collections.

Eliminating Overlapping Users: We explicitly address violation of SUTVA by eliminating users who overlap in both treatment and control groups (Jo et al. 2020). The remaining subset of over 100,000 users are exclusively in one group, thereby attenuating concerns around violation of SUTVA. The advantage of this approach is that we get a clean causal estimate for a subset of users, with the obvious downside being that the effect may not be representative of all users. Row 1 of Table 9 reports the impact of financing on quantity and prices at both platforms for non-overlapping users. We find qualitatively similar effect of financing on quantity and prices at both platforms, which show that the causal effect of lending on sales at Blur is robust to potential spillovers resulting from substitution.[21]

Minimizing Overlapping Collections: Next, we address SUTVA violation by minimizing the overlap between the treated and control group collections. Specifically, we use as control group, collections (belonging to the same categories as the original control group) that have at most 15% overlap in users with the treated collections.[22] Although we recognize that this approach does not completely eliminate concerns stemming from overlapping users, the minimal overlap should mitigate the likelihood of SUTVA being violated. Row 2 of Table 9 reports the estimates based on this control group definition. Consistent with the baseline results, we again find a positive and significant impact of financing on quantity but only at Blur, and no impact of financing on prices at either platform.[23]

[21] Appendix A6 and A7 reports estimates for this analysis using Two-Stage DiD and TWFE models, which are qualitatively similar to the main findings.

[22] We tried to include collections with only 5% or 10% overlap in users but this resulted in very few collections which made the process of computing synthetic weights unreliable.

[23] Qualitatively similar findings from Two-Stage DiD and TWFE models are reported in Appendix A6 and A7.

Table 9 Robustness of Results to Overlapping Users

	Blur		OpenSea	
	Quantity (1)	Price (2)	Quantity (3)	Price (4)
Eliminate Overlapping Users in Pre-Treatment Period	0.351**	-0.064	-0.030	-0.033
	(0.169)	(0.229)	(0.176)	(0.409)
Minimum Overlapping Control	0.544**	-0.047	0.119	0.145
	(0.266)	(0.221)	(0.223)	(0.778)
Falsification Tests				
Pseudo Treatment Group	-0.241	0.220	-0.114	-0.204
	(0.229)	(0.299)	(0.139)	(0.154)
Pseudo Treatment Week	-0.101	-0.150	-0.045	0.151
	(0.270)	(0.305)	(0.121)	(0.303)

Note. The Table reports estimates from analysis conducted to check the robustness of results to violations of SUTVA and falsification tests. The top panel of the Table reports results from two specifications, which either eliminate the overlapping users between the treatment and control groups (Row 1) or use as a control group categories that have minimal overlapping users with the treated collections (Row 2). The bottom panel of the Table reports results from two specifications, which either create pseudo-treated collections by sampling from control group collections (Row 3) or vary the timing of treatment to three weeks before the actual treatment (Row 4). All models are based on 2,499 collection-week level observations and include collection and week fixed effects. Standard errors reported in parentheses are clustered at the collection level. Asterisk indicate significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4.3. Eliminating Multi-Homing Users: While our results indicate that the introduction of financing on Blur did not generate significant spillovers in trading activity on OpenSea (Table 3), a remaining concern is that users frequently multi-home across NFT marketplaces. Even in the absence of observable platform-level spillovers, financing introduced on Blur may affect the behavior of users who actively participate on both platforms. Such cross-platform responses could potentially confound the interpretation of our triple-difference (DDD) estimates reported in Table 4. To further ensure the robustness of our findings, we construct a restricted sample that excludes transactions involving users who actively participate on both platforms during the pre-treatment period (similar to Zhang et al. 2026). Specifically, we identify multi-homing users as wallets that transact on both Blur and OpenSea and remove their transactions.

Table 10 reports the results. Reassuringly, the coefficient on *Post* × *Treated* × *Blur* in specification 1 remains positive and statistically significant and, the estimated magnitudes are comparable to those reported in Table 4. These findings suggest that our main results are not driven by cross-platform user spillovers.

‘‘‘

5.4.4. Falsification Tests: Finally, we conduct two falsification tests to evaluate whether the estimated effects could be driven by spurious correlations or treatment timing. In the first test,

Table 10 Robustness of Triple Difference Estimates to Multi-Homing Users

	Quantity (1)	Price (2)
Post × Treated × Blur	0.445** (0.197)	0.158 (0.096)

Note. This table re-estimates the triple-difference (DDD) specification reported in Table 4 after excluding all users who transact on both Blur and OpenSea during the pre-treatment period. All models are based on 4,998 collection-week observations and include the fixed effects implied by the DDD design. Standard errors reported in parentheses are clustered at the collection level. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

we randomly assign placebo treatment status to a subset of control collections and assign placebo adoption dates that mirror the distribution of actual treatment dates. In the second test, we move the treatment date three weeks prior to the actual introduction of financing. If our findings were driven by spurious correlations or anticipatory behavior, we would expect these placebo treatments to generate significant estimates.

The bottom panel of Table 9 reports the change in quantities sold on both platforms under different falsification or placebo tests (Imbens and Rubin 2015). These falsification tests check robustness of our findings to concerns around spurious correlations, either overlapping with the timing of enabling financing or between the treatment and control groups, which could confound the causal effect of financing. To conduct a falsification test for the main effect, we randomly pick collections from the control group and treat some of them as if financing were enabled for them. Specifically, we draw 13 collections from the control pool and assign them placebo treatment weeks in the same distribution as the real treatment, leaving the remaining collections never-treated.

In the absence of spurious correlation or common shocks which differentially impact treatment and control groups, we would not expect the estimate to be significant. Additionally, to check whether parallel trends are violated or if there was any effect due to financing being anticipated by users, we conduct a falsification test for the timing of adoption where we reassign the treatment date to be 3 weeks before the actual treatment date. As the adjusted treatment variable excludes the actual treatment date, we would expect the estimates to be insignificant. Consistent with this intuition, we find that the effect of financing on quantities sold is not statistically significant under both of these pseudo treatments, suggesting that the estimated effect is not driven by spurious correlation or model specification.[24] We also repeated the analysis by reassigning the treatment

[24] We get qualitatively similar results using Two-Stage DiD and TWFE models as reported in Appendix A6 and A7.

date to be 4, 5, or 8 weeks prior to the actual treatment, and again find an insignificant effect of financing which points to the results not being driven by spurious temporal correlation.[25]

6. Conclusion

Digital marketplaces are increasingly embedding financial services directly into transactions. While financing can stimulate demand by relaxing liquidity constraints, participant-provided financing may also divert capital away from trading by encouraging users to act as lenders rather than buyers. Understanding the net effect of these competing forces is therefore important for the design of tokenized marketplaces and other digital platforms that seek to integrate trading and capital provision.

In this paper, we study the staggered introduction of peer-to-peer, collateral-backed financing on Blur, a leading marketplace for non-fungible tokens. Using Synthetic Difference-in-Differences and triple-difference designs, we find that enabling financing increases transaction quantities by 49 percent while having no significant effect on average prices. The increase is driven primarily by existing users, who transact more frequently and purchase more assets after financing becomes available. Financing has little effect on the entry of new buyers or sellers, although it attracts new participants on the lending side of the marketplace. We also find that financing shifts activity toward higher-value assets, increases both listings and purchase probabilities, and generates little evidence of spillovers to competing marketplaces.

Taken together, these findings suggest that the demand-expanding effects of financing dominate any tendency for participants to reallocate capital away from trading and toward lending. Although users can simultaneously act as buyers, sellers, and lenders, introducing financing increases rather than crowds out marketplace activity. The results therefore highlight a distinct feature of tokenized marketplaces: tokenization allows the same asset to function simultaneously as a tradable good and as collateral for credit, enabling capital provision to be embedded directly within marketplace activity.

More broadly, our findings contribute to the growing literature on tokenization and digital platforms by showing that financing can become an integral marketplace design choice rather than a separate financial service. Prior work has largely examined how tokenization affects ownership, governance, coordination, and value creation. We show that tokenization also enables platforms to facilitate capital allocation among participants themselves, allowing users to provide financing without relying on traditional intermediaries. In doing so, platforms can deepen engagement, expand trading activity, and reshape the composition of assets traded within the marketplace.

[25] These results are reported in the Appendix A8.

Our study has several limitations that point to promising directions for future research. First, while we examine the marketplace consequences of financing, we do not model how lenders determine loan terms such as interest rates or loan-to-value ratios. Understanding how lenders price risk in tokenized lending markets remains an important open question. Second, our analysis focuses on short-run marketplace outcomes. Future work could examine how financing influences longer-run outcomes such as creator incentives, asset creation, platform growth, and market structure.

As tokenization extends beyond digital collectibles to encompass real estate, securities, commodities, and other asset classes, platforms are increasingly serving not only as marketplaces for exchange but also as mechanisms for allocating capital. Understanding how embedded financing shapes participation, trade, and competition is therefore important for platform designers.

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7. Appendix

7.1. Other Blur Incentives:

Blur employed a few incentive mechanisms during our sample period. Understanding these mechanisms is important to rule out alternative explanations for our findings.

Trading and Listing Points. Throughout 2023, Blur rewarded users with points for listing NFTs at competitive prices, placing bids on collections, and completing trades. These points were accumulated across "seasons" and converted into $BLUR token airdrops at season conclusions. Season 1 concluded in February 2023, and Season 2 ran through much of our sample period. However, these incentives ensure they do not threaten our identification strategy. First, trading and listing points applied to all collections on Blur, both treated and control, and both in the pre- and post-treatment period. Because these incentives did not vary at the collection level, over time, their effects are absorbed by the week fixed effects in our specification. Second, the timing of these incentive programs was independent of the staggered rollout of financing across collections. Third, our comparison of treated collections on Blur versus the same collections on OpenSea (where no Blur points applied) provides an additional check: if Blur's general incentive programs were driving our results, we would expect to see effects across all Blur collections, not specifically those with financing enabled.

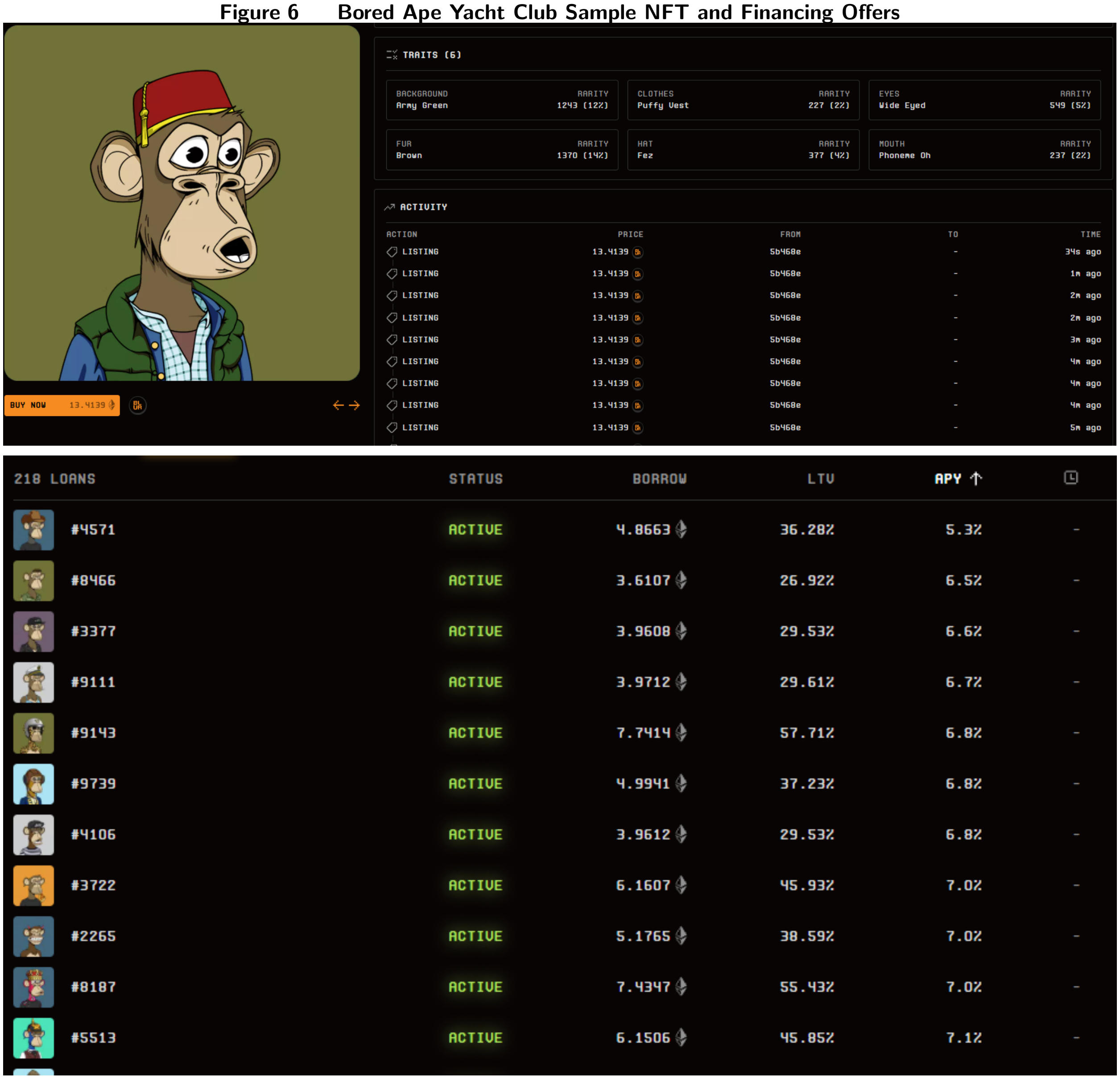
Lending points were introduced alongside the financing mechanism and applied exclusively to financing-enabled collections. These points incentivized the supply of loan offers rather than trading activity per se. Our identification, therefore, isolates the effect of financing availability, net of platform-wide incentive effects common across collections.

7.2. Other Platform Dynamics:

Several other platform-level changes occurred during our sample period. First, both Blur and OpenSea adjusted their royalty policies. Blur enforced a minimum 0.5% creator royalty throughout 2023, while OpenSea moved to optional royalties (0.5% minimum) in February 2023 and retired its royalty enforcement tool in August 2023. Second, in July 2023, Blur reduced gas fees by approximately 50% with the launch of Blur v2. Third, OpenSea responded to Blur's competitive pressure by temporarily eliminating its 2.5% trading fee in February 2023. Importantly, none of these changes varied between treated and control collections. Royalty and fee policies applied uniformly across all collections on each platform, and OpenSea did not introduce any financing mechanism during our sample period. These platform-wide effects are therefore absorbed by our time-fixed effects and do not confound our identification of the financing treatment effect. Our estimates capture the differential impact of financing availability on treated collections relative to control collections, net of any common shocks affecting all collections on each platform.

7.3. Example of NFT and Loan Offers

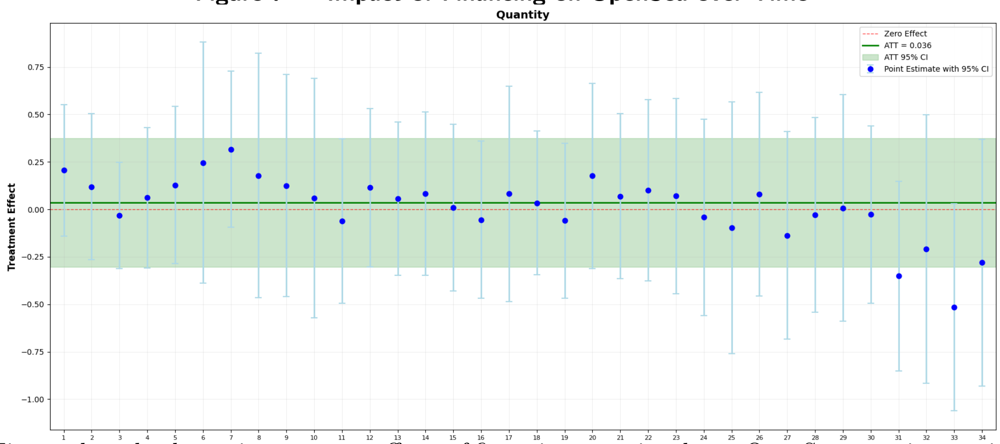
Figure 6 Bored Ape Yacht Club Sample NFT and Financing Offers



Note. The top panel of the Figure shows a screenshot of an NFT from the Bored Ape Yacht Club (BAYC). The screenshot details NFT attributes such as fur color, eye color, hat, clothes, etc. The bottom panel of the Figure shows how loan options for any NFT are provided to users on Blur.

7.4. Dynamic Spillover Effects at OpenSea

Figure 7 Impact of Financing on OpenSea over Time



Note. The Figure plots the dynamic treatment effects of financing on unit sales at OpenSea over time, estimated using the Synthetic DiD model. Each point represents the period-specific treatment effect estimate, and the vertical bars indicate the 95% confidence interval. The horizontal green line denotes the average treatment effect on the treated (ATT), and the dashed line indicates the zero-effect benchmark.

7.5. Descriptive Statistics of All Collections

Table A1 Financing Enabled Collections

Collection	Financing Week(2023)	Avg. Unit Sold	SD Quantity	Collection Size	Active Users
Azuki	18	613.68	703.87	10,000	10768
Beanzofficial	21	1425.05	1908.68	19,950	24370
Bored-ape-kennel-club	20	442.13	325.55	9,602	8756
Boredapeyachtclub	18	237.37	154.77	9,998	5610
Clonex	21	612.86	617.87	19,764	11222
Kanpai-pandas	21	206.70	128.07	7,998	5313
Milady	18	616.90	533.01	9,733	12185
Mutant-ape-yacht-club	20	810.70	416.71	19,493	15446
Otherdeed	21	1597.92	2634.72	45,920	23345
Pudgypenguins	21	500.01	382.162	8,888	9567
Remilio-babies	21	673.66	543.84	9,172	12967
Moonbirds	48	547.82	629.60	9,999	9111
Lilpudgys	49	927.35	806.88	21,653	19310
Mean		708.63	752.75	15551.54	12920.77

Note. Each row of the Table provides statistics corresponding to each of the 13 collections for which financing was enabled. The columns report the average and standard deviation of sales, collection sizes, and the number of active users. The average statistics across all treated collections are reported in the bottom row.

Table A2 Other PFPs Collections

Collection	Avg. Unit Sale	SD	Collection Size	Active Users
Doodles	499.82	559.2602	9,998	8627
Meebits	298.61	272.7064	20,000	6024

Note. Each row of the Table provides statistics for the PFPs collections included in the analysis. The columns report the average and standard deviation of sales, collection sizes, and the number of active users.

Table A3 Top 11 Art Collections by Trading Volume

Collection	Avg. Unit Sale	SD	Collection Size	Active Users
CrypToadz by GREMPLIN	79.647	94.237	7,022	2,243
VV-checks-originals	604.689	1,180.491	4,922	10,545
10ktf	149.177	125.132	18,100	4,101
The Dooplicator	217.078	394.848	9,375	5,098
Renga	535.569	733.818	8,344	10,899
Terraforms	125.588	110.967	9,911	3,278
Adam Bomb Squad	66.431	45.855	24,936	2,026
CreatureWorld	60.706	46.312	9,999	1,779
DeadFellaz	115.451	82.213	9,999	3,778
Friendship Bracelets by Alexis Andre	469.961	1,128.233	38,651	7,687
OnChainMonkey	99.863	79.798	10,000	3,166

Note. Each row of the Table provides statistics for the Art collections included in the analysis. The columns report the average and standard deviation of sales, collection sizes, and the number of active users.

Table A4 Top 13 Membership Collections by Trading Volume

Collection	Avg. Unit Sale	SD	Collection Size	Active Users
SomethingToken	38.020	24.001	9,925	856
Proof-collective	20.500	22.472	1,000	817
Psychedelics Anonymous Genesis	94.549	60.374	9,593	2,935
VaynerSports Pass VSP	47.941	32.901	15,554	1,229
Writers Room	32.280	43.671	6,925	934
Jrny Club Official	43.235	16.587	5,437	1,358
Premint Collector	94.471	80.938	10,000	3,404
Hausphases by Fabrik	29.176	29.311	7,773	763
Crypto Bull Society	15.941	13.377	7,777	513
Neo Tokyo Citizens	9.440	4.510	3,227	215
GEN.ART Membership	9.116	14.390	5,092	267
LinksDAO	42.490	32.262	9,090	1,382
Flyfish Club	5.064	3.313	1,631	224

Note. Each row of the Table provides statistics for the Membership collections included in the analysis. The columns report the average and standard deviation of sales, collection sizes, and the number of active users.

Table A5 Top 10 Gaming Collections by Trading Volume

Collection	Avg. Unit Sale	SD	Collection Size	Active Users
Rumble Kong League	43.590	28.930	10,000	1,271
My Pet Hooligan	98.780	59.470	8,888	3,034
CyberKongz (Babies)	48.980	30.340	4,000	1,642
Wolf Game	113.250	122.050	13,617	3,119
Worldwide Webb Land	106.510	211.280	9,508	2,573
VOX Collectibles: Town Star	18.370	11.050	8,888	545
Metroverse Genesis	39.720	60.650	9,923	876
Pixelmon - Generation 1	224.750	273.620	12,883	4,395
Treeverse Plots	93.020	79.370	10,400	2,218
Impostors Genesis Aliens	85.140	43.480	10,420	2,853

Note. Each row of the Table provides statistics for the Gaming collections included in the analysis. The columns report the average and standard deviation of sales, collection sizes, and the number of active users.

7.6. Robustness to Overlapping Users with Alternate Specifications

Table A6 Robustness of Results to Overlapping Users - Two-Stage DiD

	Blur		OpenSea	
	Quantity (1)	Price (2)	Quantity (3)	Price (4)
Eliminating Overlap in Users				
Eliminate Overlapping Users	0.485**	0.185	-0.007	0.229
	(0.222)	(0.220)	(0.140)	(0.219)
Minimum Overlapping Control	0.395*	0.244	0.082	0.265
	(0.231)	(0.243)	(0.165)	(0.223)
Falsification Tests				
Pseudo Treatment Group	-0.355	0.040	-0.164	0.014
	(0.251)	(0.239)	(0.162)	(0.213)
Pseudo Treatment Week	-0.203	-0.039	-0.081	0.084
	(0.201)	(0.198)	(0.157)	(0.287)

Note. The Table reports estimates from analysis conducted to check the robustness of results to violations of SUTVA and falsification tests using the **Two-Stage DiD** model. The top panel of the Table reports results from two specifications, which either eliminate the overlapping users between the treatment and control groups (Row 1) or use as a control group categories that have minimal overlapping users with the treated collections (Row 2). The bottom panel of the Table reports results from two specifications, which either create pseudo-treated collections by sampling from control group collections (Row 3) or vary the timing of treatment to three weeks before the actual treatment (Row 4). All models are based on 2,499 collection-week level observations and include collection and week fixed effects. Standard errors reported in parentheses are clustered at the collection level. Asterisk indicate significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7 Robustness of Results to Overlapping Users - TWFE Model

	Blur		OpenSea	
	Quantity (1)	Price (2)	Quantity (3)	Price (4)
Eliminating Overlap in Users				
Eliminate Overlapping Users	0.489***	0.087	0.013	0.174
	(0.184)	(0.213)	(0.184)	(0.445)
Minimum Overlapping Control	0.382*	-0.044	0.139	0.151
	(0.225)	(0.260)	(0.187)	(2.796)
Falsification Tests				
Pseudo Treatment Group	-0.252	0.067	-0.094	0.001
	(0.197)	(0.288)	(0.169)	(0.264)
Pseudo Treatment Week	-0.135	-0.030	-0.016	0.062
	(0.231)	(0.249)	(0.148)	(0.346)

Note. The Table reports estimates from analysis conducted to check the robustness of results to violations of SUTVA and falsification tests using the **Two-Way Fixed Effects** model. The top panel of the Table reports results from two specifications, which either eliminate the overlapping users between the treatment and control groups (Row 1) or use as a control group categories that have minimal overlapping users with the treated collections (Row 2). The bottom panel of the Table reports results from two specifications, which either create pseudo-treated collections by sampling from control group collections (Row 3) or vary the timing of treatment to three weeks before the actual treatment (Row 4). All models are based on 2,499 collection-week level observations and include collection and week fixed effects. Standard errors reported in parentheses are clustered at the collection level. Asterisk indicate significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7.7. Falsification Tests with Different Weeks

Table A8 Falsification test results assuming treatment occurred 4, 5, and 8 weeks prior to the intervention

	Blur (1)	OpenSea (2)
Quantity (week 4)	-0.135	-0.064
	(0.232)	(0.171)
Quantity (week 5)	-0.124	-0.027
	(0.255)	(0.161)
Quantity (week 8)	-0.157	-0.050
	(0.172)	(0.175)

Note. The Table reports estimates from Falsification tests, which vary the timing of the treatment. Each row presents estimates from a model that assumes that the treatment happened n weeks before the actual treatment, as indicated in the rows. All models are based on 1,836 collection-week level observations and include collection and week fixed effects. Standard errors reported in parentheses are clustered at the collection level. Asterisk indicate significance: * $p<0.10$, ** $p<0.05$, *** $p<0.01$.